10. Search and rescue — rubble, fire, floods and lost people

Oracle OS guide. Based on FEMA CERT, INSARAG (UN), the US&R Field Operations Guide and RNLI (illustrated in the library: "FEMA CERT Basic Training", "INSARAG Guidelines vol. II", "US Urban SAR Field Operations Guide", "NWCG Incident Response Pocket Guide", "PHMSA Emergency Response Guidebook").

Golden rule

Don't become another casualty. Many would-be rescuers die trying to save someone. Assess first, act second, always keep an exit.

Size up the scene (before going in)

- What happened? How many people? Where could they be?
- Hazards: fire, smoke, gas (smell → get out, don't touch switches), electricity, water, chemicals (orange plates with numbers: look them up in the "Emergency Response Guidebook"), collapsing structures, aftershocks.
- Resources: who is with you, tools, light, radio, remaining daylight.
- Shut off gas and electricity at the meter if safe.

Damaged buildings (earthquake, explosion, collapse)

- **Light damage** (cracks, glass): enter carefully.
- **Moderate damage** (cracked walls, damaged stairs): only quick entry for visible casualties.
- **Heavy damage** (fallen walls/roof, leaning, off foundation): **do not enter**. Mark it, cordon it, wait for professionals.
- Work in pairs, with helmet/gloves/dust mask and sturdy shoes. One person stays outside and knows where you are.

Finding casualties

- **Call out and listen**: "If you can hear me, knock 3 times!" Then **total silence** for 1–2 minutes. Repeat from several points and triangulate.
- Search void spaces: under tables, along interior walls, stairwells, corners, vehicles.
- Surface and lightly trapped casualties first (most survivors are pulled out in the first hours by neighbours).
- **Marking** at the entrance (spray or chalk): an X — left: team; top: date and time; right: hazards; bottom: casualties (alive/dead). Prevents repeated searches.

Freeing someone trapped

- **Lever and crib**: lift a little with a lever (bar, strong plank on a fulcrum), slide in a block; **lift an inch, crib an inch**. Never leave a load unsupported.
- Don't pull beams that hold other parts up. Remove debris by hand, passing it along a chain, top down.

- **Crush syndrome**: someone with a limb crushed for more than 1–2 hours can deteriorate suddenly when freed. Give water if conscious, get medical help, monitor; don't leave them alone.
- Protect the casualty from dust (cloth over face) and cold during extraction.

Moving casualties

- Only for immediate danger. Suspected spinal injury: move as a unit, head in line.
- **Clothing drag** (by the shoulders of the top, head cradled on your forearms) or **blanket drag**.
- **Two-person carry**: hand seat, or one at the armpits and one at the knees.
- **Improvised stretcher**: two 2 m poles + coats with sleeves turned inside, or a blanket folded in thirds around the poles. Test with a healthy adult first.

Fire and smoke

- Crawl below smoke; feel the door with the back of your hand before opening — hot, don't open.
- Close doors behind you (slows the fire). Trapped: close the door, seal gaps with wet cloth, signal from the window.
- Clothes on fire: **stop, drop, roll**. Cool burns 20 min with water.
- Carbon monoxide (generators, barbecues indoors): headache, nausea, drowsiness in several people → fresh air now.
- Wildfire: move to already-burnt ground, bare ground, water or a wide road; never run uphill ahead of the fire.

Water — floods, rivers, ice

- **Reach, throw, row, go** — in that order. Reach with a stick, rope, clothing; throw something that floats (empty jerrycan, ball, life jacket). Only go in trained and with buoyancy.
- Moving water: **15 cm can knock an adult over, 30 cm can float a car**. "Turn around, don't drown."
- Ice: don't run to the casualty. Lie flat, spread your weight, push out a pole/ladder; pull them to the bank.
- See the "Water, rivers and sea" guide for currents, crossings and sea survival.

Signals to be found

- **3 of anything** = distress: 3 whistle blasts, 3 fires in a triangle, 3 flashes. **SOS** in Morse: · · · — — — · · ·
- A whistle carries much further than a voice and uses less energy.
- **Ground-to-air signals** (large, 3 m+, contrasting): **V** = need help; **X** = need medical help; **N** = no; **Y** = yes; **arrow** = went this way.
- Mirror/shiny surface to flash sunlight at aircraft; bright colours; smoke by day (green leaves), fire by night.

- Radio: marine VHF channel 16; PMR446 channel 8 (code 0) and channel 1 are common emergency channels; say who, where, what, how many.

Lost person

- If it's you: **STOP** — Stop, Think, Observe, Plan. Stay put if someone knows your route; shelter, signal.
- Children: teach them to "hug a tree" and stay still.
- Searching: start at the last seen point; check likely routes, junctions, water, shelters. Line search with people within sight of each other, calling and pausing to listen. Mark areas already searched.

Reporting an emergency (radio or phone)

1 Who you are and where you're calling from. 2. **Where** (Oracle coordinates: Emergency tab). 3. What happened. 4. How many casualties and their state. 5. Hazards. 6. What you've done and what you need. Don't hang up first.

Household rescue kit

Helmet or hard hat, work gloves, goggles, FFP2 masks, head torch + batteries, whistle, crowbar, hammer, saw, gas/water shut-off key, 15–30 m rope, barrier tape, chalk/spray, markers, blanket, first aid kit.